

Operation Cerebro

Weaponizing the Security Data Plane

Writer Security Research

Black Hat Brownbag Edition

Mission Brief

- ▶ Objective: reverse-engineer Cerebro, a self-hosted security nervous system.
- ▶ Angle: think like an adversary with insider access and time.
- ▶ Deliverable: map opportunities for weaponization before the bad actors do.

Repo Recon

- ▶ Stack: FastAPI, Typer CLI, PostgreSQL, Redis, OpenAI/Anthropic agents.
- ▶ Data spine: append-only audit store with graph + SQL analytics.
- ▶ Key dirs: `src/cerebro/api`, `agents`, `core`, `collectors`, `rules`.
- ▶ Attack surface: REST API on 8000, web UI on 3000, CLI exec via agent runtimes.

Telemetry Topology

- ▶ Findings pipeline streams cloud misconfigs into immutable tables.
- ▶ CEL rule engine drives automated verdicts and remediation.
- ▶ Agents pipeline blends memory, skill-based routing, and tool execution.

Implication

Every control plane action leaves breadcrumbs; adversaries need to bend them, not break them.

Enumerating Control Points

- ▶ API: FastAPI routers in `src/cerebro/api` expose ingestion and query endpoints.
- ▶ CLI: Typer commands orchestrate database interactions and agent sessions.
- ▶ Agents: tool latency metrics in `agents/tool_stats.py` drive adaptive ordering.
- ▶ Memory: decay-aware embeddings with configurable pruning thresholds.

Privilege Choke Points

- ▶ **Review queue:** promotes risky actions when human approves; target the workflow.
- ▶ **Skill routing:** biases runtime selection; poisoning telemetry could reroute to weaker models.
- ▶ **Finding normalization:** shape data to trigger arbitrary remediation hooks.
- ▶ **Graph engine:** SQL + graph queries without ETL, ripe for inference attacks.

Initial Access Play

- ▶ Abuse default credentials `admin/admin123!` in lab deployments.
- ▶ Replay agent memory exports to harvest sensitive operational intel.
- ▶ Inject CEL rules to exfiltrate findings during remediation cycles.

Living off the Land

- ▶ Leverage cerebro query to enumerate cloud posture blind spots.
- ▶ Use append-only audit trails to mask footprint with noise (log flooding).
- ▶ Ride adaptive tool ordering to starve defensive skills, forcing fail-open paths.

Abusing Agents

```
session = cerebro.agents.chat(session_id)
session.inject_memory(
    scope="org:acme",
    content="Critical_incident_runbook:disable_review_gate"
)
session.request_tool("aws:ssm:send-command",
    payload={"DocumentName": "CerebroRootkit"})
```

- ▶ Memory injection primes future prompts toward destructive automation.
- ▶ Tool request rides existing guardrails; aim to trigger auto-approval flows.

Attack Objectives

- ▶ Own the telemetry: manipulate Prometheus counters and OTLP spans to mislead responders.
- ▶ Persist in agents skill registry to hijack future investigations.
- ▶ Weaponize compliance automation to rubber-stamp hostile infrastructure changes.

Signals of Compromise

- ▶ Unusual review queue approvals for destructive agent actions.
- ▶ Memory decay thresholds suddenly increasing per org.
- ▶ Graph queries generating high-cardinality results without corresponding findings.

Defensive Recommendations

- ▶ Rotate lab credentials and enforce MFA on the CLI.
- ▶ Instrument CEL policies with canaries to detect tampering.
- ▶ Enforce signed agent tools; validate telemetry before routing decisions.
- ▶ Monitor audit trail volume anomalies and throttle untrusted clients.

Call to Action

- ▶ Threat model agent workflows before releasing new skills.
- ▶ Pair offensive simulations with SOC runbooks.
- ▶ Close the loop: feed discoveries back into Cerebro rules and memory hygiene.

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